

SecureMailScope

SIH26159 — AI-Assisted Cryptographic Security Posture Assessment

Scan: SMS-DEMO-20260929-195736-9437 | Mode: CONTROLLED_DEMO
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Overall posture: 0/100 — CRITICAL RISK

Sessions: 6 | Secure/Encrypted sessions: 5 | Findings: 15 | Critical: 1 | High: 8 | Medium: 5 | Low: 1

Session Inventory

Session	Protocol	TLS	Cipher	Certificate	Packets
smtp-001	SMTP	TLS 1.3	TLS_AES_256_GCM_SHA384	valid	42-118
smtp-002	SMTP	TLS 1.1	TLS_RSA_WITH_3DES_EDE_CBC_SHA	expired	201-319
imap-003	IMAP	TLS 1.2	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	valid	441-512
pop3-004	POP3	PLAINTEXT	NONE	not-observed	611-642
smtp-005	SMTP	TLS 1.2	TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384	valid	733-812
imap-006	IMAP	TLS 1.0	TLS_RSA_WITH_RC4_128_SHA	expired	905-988

Findings

ID	Severity	Session	Finding	Evidence
TLS-001	HIGH	smtp-002	Deprecated TLS version negotiated	Packets 201-319: negotiated TLS version = TLS 1.1
CRYPTO-002	HIGH	smtp-002	Weak / deprecated cipher suite	Packets 201-319: cipher suite = TLS_RSA_WITH_3DES_EDE_CBC_SHA
CRYPTO-003	MEDIUM	smtp-002	Forward secrecy not observed	Packets 201-319: key exchange = RSA
CERT-004	HIGH	smtp-002	Expired X.509 certificate	Packets 201-319: certificate validity indicates expiry
CERT-005	MEDIUM	smtp-002	SHA-1 certificate signature observed	Packets 201-319: certificate signature = SHA-1 with RSA
CERT-006	MEDIUM	imap-003	SHA-1 certificate signature observed	Packets 441-512: certificate signature = SHA-1 with RSA
CERT-007	LOW	imap-003	Certificate expires soon	Packets 441-512: certificate expires within the preferred 30-day operation
TLS-008	CRITICAL	pop3-004	Email session observed without TLS protection	Packets 611-642: email payload observed before any TLS handshake
TLS-009	HIGH	imap-006	Deprecated TLS version negotiated	Packets 905-988: negotiated TLS version = TLS 1.0
CRYPTO-010	HIGH	imap-006	Weak / deprecated cipher suite	Packets 905-988: cipher suite = TLS_RSA_WITH_RC4_128_SHA
CRYPTO-011	MEDIUM	imap-006	Forward secrecy not observed	Packets 905-988: key exchange = RSA
CERT-012	HIGH	imap-006	Expired X.509 certificate	Packets 905-988: certificate validity indicates expiry
CERT-013	HIGH	imap-006	Weak public-key size	Packets 905-988: public key = RSA 1024
CERT-014	MEDIUM	imap-006	SHA-1 certificate signature observed	Packets 905-988: certificate signature = SHA-1 with RSA
CERT-015	HIGH	imap-006	Certificate identity mismatch	Packets 905-988: observed server name not covered by certificate SAN

AI-Assisted Insights

- The evidence contains 1 critical finding(s); these should be addressed first.
- Certificate controls are the dominant source of observed cryptographic risk in this capture.
- Isolation Forest highlights imap-006 as the highest-anomaly session with score 1.0.
- Plaintext email access/submission is a direct transport-security concern and creates the strongest evidence-backed exposure in the demonstration capture.
- Risk conclusions are derived from observed protocol/cryptographic evidence; anomaly scores identify unusual profiles and do not by themselves prove compromise.

Recommendations

- Require TLS-protected email access/submission and disable plaintext access where policy permits.
- Disable TLS 1.0/1.1 and enforce TLS 1.2+; prefer TLS 1.3.
- Remove RC4/3DES and permit modern AEAD suites such as AES-GCM or ChaCha20-Poly1305.
- Renew the certificate and validate the complete trust chain before deployment.
- Replace 1024-bit RSA material with a modern key size/algorithm.
- Deploy a certificate whose SAN/CN covers the observed mail service hostname.
- Prefer ephemeral ECDHE/X25519 key exchange to provide forward secrecy.
- Re-issue the certificate using a modern signature algorithm such as SHA-256 or stronger.

Integrity Evidence

Final evidence hash: 293ac626ea5c6e292cd303a0189cfa5eaae0d9f917fa9aef04fac0566b742d65

Final audit record hash: ca270d6a2a2ff748a58e0cd2241c658336bd656c8fd48013ce8838dd84b5592b

The current prototype uses a chained SHA-256 audit ledger. A production deployment can anchor the final evidence hash to a permissioned ledger.